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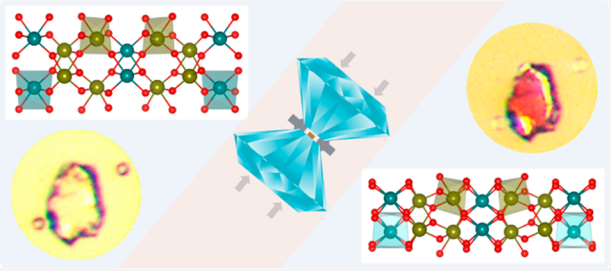
Pressure-Induced Phase Transition in $Mn(Ta,Nb)_2O_6$: An Experimental Investigation and First-Principle Study

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ABSTRACT: The high-pressure and high-temperature behaviors of manganotantalite $Mn(Ta,Nb)_2O_6$ have been investigated by single-crystal X-ray diffraction and Raman spectroscopy combined with diamond anvil cell technique, as well as first-principle calculations. A pressure-induced reversible phase transition of manganotantalite occurs at 9.5 GPa and room temperature, accompanied by a large volume collapse (~7.0%) and drastic color change from brownish-yellow to red. The space groups of low-pressure (LP) and high-pressure (HP) phases are the same (*Pbcn*), but the coordination numbers of Mn increase from six to eight and Ta increase from six to seven, respectively. The band gap becomes narrow from 2.37 to 1.59 eV. We determined the *P-T* phase diagram of manganotantalite with a positive Clapeyron slope of $dP/dT = 0.0073$ GPa/K. The *P-V* data were fitted to a second-order Birch−Murnaghan equation of state with $B_0 = 149(4)$ GPa for the LP phase and $B_0 = 188(3)$ GPa for the HP phase. The isothermal Grüneisen parameters were determined to be 0.23~2.03 of the LP phase and 0.59~0.86 of the HP phase for Raman modes. High-pressure behaviors of $Mn(Ta,Nb)_2O_6$ indicate that this kind of material is a potential effective pressure sensor to monitor pressure change or warn pressure abnormality.



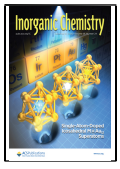
1. INTRODUCTION

The families of oxide compounds with the general formula AB_2O_6 (A-Fe, Mn, Mg, Zn, Ca, Co, Ni; B-Ta, Nb, V, Pb) have attracted much interest because of their important use in technological applications.[1−3] The high permittivity and low dielectric loss at microwave frequencies make these materials suitable for use as resonators in the fields of satellite communications, supercapacitors in electric cars, ceramics in microwave technology, and many other electric devices.[4−9] The AB_2O_6 compounds are also used as important intermediates for the synthesis of perovskite structure related materials.[5] Among AB_2O_6 compounds, $(Mn,Fe)(Nb,Ta)_2O_6$ with A and B sites mainly occupied by Mn^{2+} or Fe^{2+} and Ta^{5+} or Nb^{5+}, respectively, deserves special attention due to its potential applications.[10]

In the natural state, $(Mn,Fe)(Nb,Ta)_2O_6$ compounds are found in granitic pegmatites or highly differentiated granite bodies enriched in volatile elements.[11] The $(Mn,Fe)(Nb,Ta)_2O_6$ family of compounds encompass two different solid solution series, named tantalite-columbite with orthorhombic structure and tapiolite-mossite with tetragonal structure, the compositional fields were presented in literature, of which the manganotantalite $Mn(Ta,Nb)_2O_6$ belongs to tantalite−columbite series.[12,13] They are the most important economic minerals containing base metals: Ta and Nb. In a stable state, manganotantalite shows a columbite-type structure that crystallizes in orthorhombic symmetry with space group of *Pbcn*, in which edge-sharing $[AO_6]$ and $[BO_6]$ octahedra chains run parallel along the *c*-axis.[13,14] There are other four synthetic polymorphs of $MnTa_2O_6$ formed at different temperatures and pressures, which have trirutile, wolframite, wodginite, and modulated α-PbO_2 structures, respectively.[15] Previous studies have focused on the properties of $(Mn,Fe)(Nb,Ta)_2O_6$ compounds at ambient and high-temperature conditions.[16−18] It is reported that the changes in thermal expansion coefficients of unit cell parameters revealed anisotropic expansion behaviors of $FeNb_2O_6$, $FeTa_2O_6$, and $MnTa_2O_6$ with temperature.[10,19,20] The thermodynamic research shows that $FeNb_2O_6$ undergoes a phase transition from orthorhombic to tetragonal structure at 1630 K.[18]

High pressure has been employed as an efficient tool for improving the understanding of the properties of functional materials. For instance, structural transition and lattice distortions induced by pressure in α-$Co[N(CN)_2]_2$ alter the superexchange interactions and correlate with the ferromagnetic-antiferromagnetic crossover.[21] The property of pressure-induced spin-crossover of iron compound with a reversible color change makes it a suitable candidate as pressure sensor.[22]

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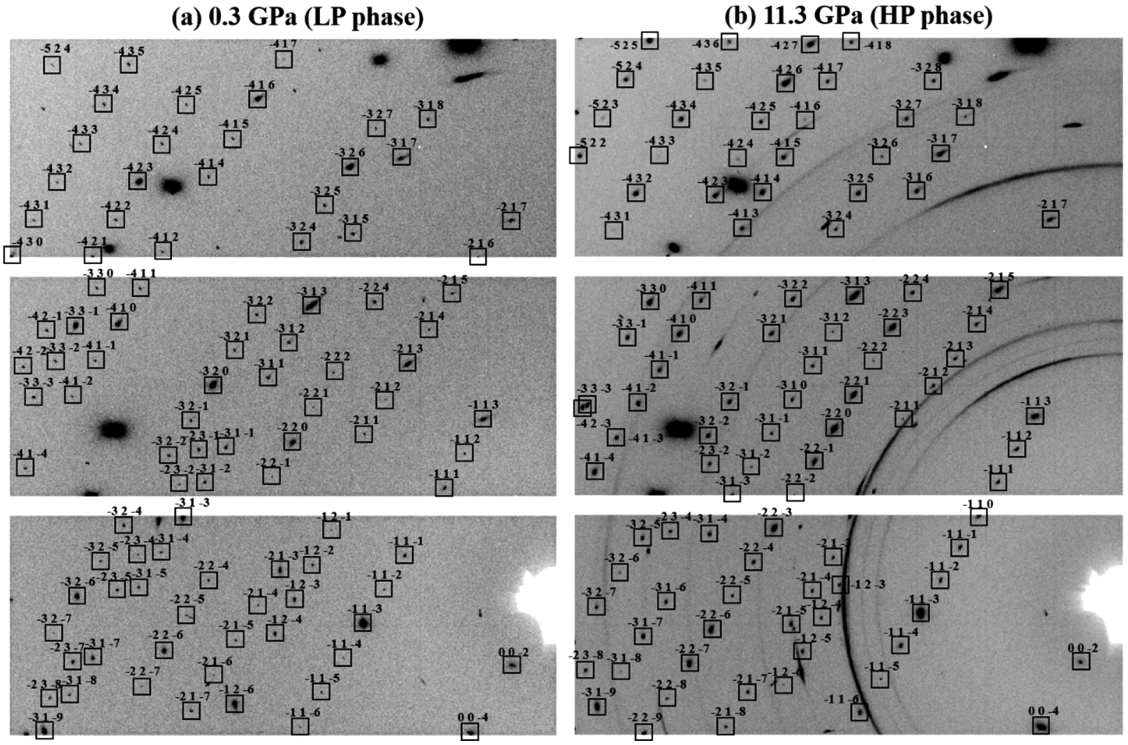


Figure 1. Examples of the two-dimensional XRD patterns of manganotantalite (a) at 0.3 and (b) 11.3 GPa. Indices are given for manganotantalite reflections (underlined by boxes). Full patterns are given in Figure S1 in the Supporting Information. Large black spots are due to diamond reflections. The diffraction rings are due to gasket (Re) reflections.

The phase transitions induced by pressure cause changes in electrical resistivity of three candidate materials ($InVO_4$, $InNbO_4$ and $InTaO_4$) for hydrogen production.[23] However, high pressure studies on $(Mn,Fe)(Nb,Ta)_2O_6$ compounds are limited. The compressional behaviors of $(Fe,Mn)(Nb,Ta)_2O_6$ columbite up to 7 GPa at room temperature reveal that the substitution of Mn for Fe and cation disordering could decrease its bulk modulus.[24,25] A pressure-induced phase transition was observed on the endmember $MnNb_2O_6$ at 12 GPa and room temperature, but this transition is supposed to be partly reversible and there is no crystal structure analysis of the new phase.[14] Further, the knowledge of fundamental crystal chemistry information and properties variation in connection with the prospects of compound manganotantalite $Mn(Ta,Nb)_2O_6$ under high-pressure and high-temperature (HPHT) conditions without changing the composition is curious and needs to be understood.

In this work, we report a comprehensive investigation on the behaviors of manganotantalite under HPHT conditions by X-ray diffraction (XRD), Raman spectroscopy and theoretical calculations. We focused on the compressibility, structural stability, *P-T* phase diagram, and potential properties. The combined results will help us to understand the behaviors of the ternary oxides with columbite structure.

2. MATERIALS AND METHODS

A natural gem-quality, transparent and brownish-yellow single crystal of manganotantalite was used for the present study. Sample characterizations were performed using electron microprobe analysis and single-crystal XRD at ambient conditions. Analysis of the chemical compositions was conducted using JEOL JXA-8100 electron probe microanalyzer at China University of Geoscience, operating with an acceleration voltage of 15 kV and a beam current of 20 nA. The chemical formula was determined to be $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ on the basis of six oxygen atoms per formula. The crystal structure was determined to be an orthorhombic phase (space group *Pbcn* and $Z = 4$) with lattice constants of $a = 14.4162(8)$ Å, $b = 5.7572(3)$ Å, $c = 5.0850(2)$ Å, and $V = 422.04(4)$ Å^3 by X-ray single-crystal diffractometer equipped with Mo Kα radiation at Huazhong University of Science and Technology.

In situ high-pressure single-crystal XRD experiments were carried out at beamline 13-BMC with the wavelength of 0.43409 Å at Advanced Photon Source (APS), Argonne National Laboratory (ANL).[26] A short symmetry-type diamond anvil cell (DAC) fitted with Boehler-type diamond anvils (400-μm flat culets) was employed to achieve high pressure and 60° opening.[27] The preindented rhenium gasket of 35 μm thickness has been drilled with a hole of 260 μm diameter as the sample chamber. Platinum was used for pressure calibration and neon was employed as pressure transmitting medium using the COMPRES/GSECARS gas-loading system.[28,29] Stepped exposures were collected in a rotation range from −30° to 30° with a 1° step, and the exposure time was 1 s per degree. XRD patterns were recorded with Pilatus3 1 M detector and experimental data were analyzed using Bruker APEX3 software. In situ Raman spectra at different pressures and temperatures were collected using a Horiba LabRAM HR Evolution micro-Raman spectrometer with a 600 lines/mm grating and 532 nm laser at China University of Geoscience. An SLM Plan 50× long working distance Olympus microscope objective was adopted to focus the laser beam and collect the scattered light. A BX90-type DAC equipped with low fluorescence diamond anvils (400 μm flat culets) was used for Raman experiments.[30] The gasket was prepared in the same manner as XRD experiments. For the high-pressure Raman experiments, ruby and neon were employed as the pressure marker and pressure transmitting medium.[31] For the HPHT Raman experiments, alumina ceramic coiled by one platinum wire of 200-μm-diameter was used as a ring heater and its resistance was approximately 2 Ω; K-type thermocouple attached to one of the diamond anvils and closed to the sample chamber was used to determine the temperatures; cubic boron nitride and NaCl were used as pressure marker and pressure transmitting medium, respectively.[32]

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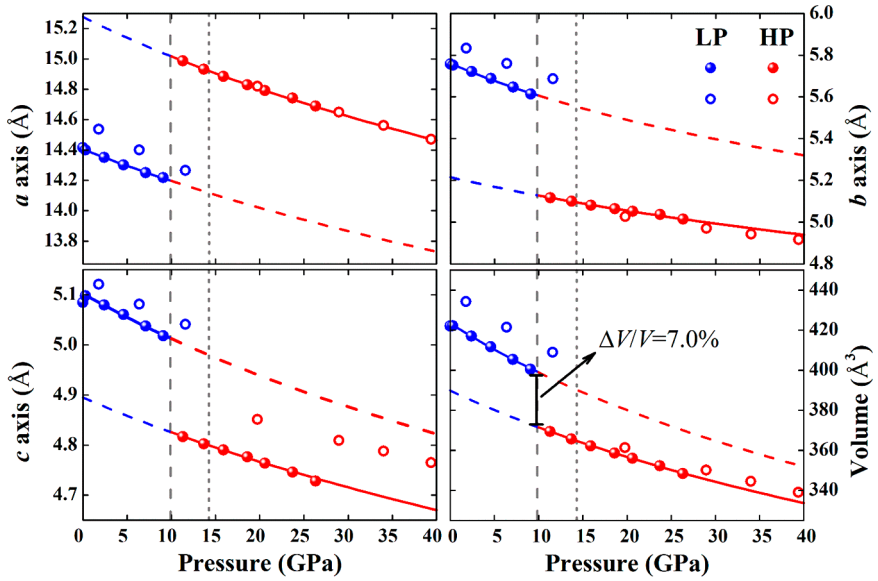


Figure 2. Lattice parameters plotted as a function of pressure. The blue and red balls represent lattice parameters of manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ at different pressures, the blue and red solid or dashed lines represent fitting curves using second-order BM EoS. The blue and red circles represent lattice parameters of manganotantalite $MnTa_2O_6$ using theoretical calculations. The gray dashed and short dashed lines represent phase transition pressure based on experiments and theoretical calculations, respectively.

First-principle calculations were performed on the endmember of manganotantalite $MnTa_2O_6$ based on the density functional theory with the projected augmented wave (PAW) method implemented in Vienna ab initio simulation package (VASP).[33,34] We chose the Perdew-Burke-Ernzerhof (PBE) version of the generalized gradient approximations (GGA) to treat the exchange correlation potential.[35] The kinetic energy cutoff was set to 800 eV and the energy convergence criterion for the electronic self-consistent calculation was 10^{-5} eV. A DFT+U method was introduced in the simulation to correctly describe the strong electronic correlation, meanwhile we applied U = 4 eV and J = 0.7 eV for Mn in all GGA+U calculations.[36] The spin-polarization of Mn without spin−orbit coupling in $MnTa_2O_6$ was included and the antiferromagnetic arrangement of Mn was considered in this study. The k-points grids were set as 2 × 6 × 6 and 4 × 12 × 12 in structural relaxation calculations and band gap simulations, respectively. The calculations were carried out with constraint the space group of *Pbcn* at all pressures in this study.

3. RESULTS AND DISCUSSION

3.1. Single-Crystal X-ray Diffraction. Natural manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ was compressed up to 26.3 GPa at room temperature. Examples of the recorded XRD patterns at different pressures are presented in Figures 1 and S1. Basing on the reconstructed 3-D lattice in the reciprocal space,[26] the reflections are indexed by orthorhombic symmetry in the range of experimental pressures. Accordingly, the obtained lattice parameters were summarized in Table S1 and plotted in Figure 2. A discontinuity at approximately 9.5 GPa has been observed on the unit-cell volumes, indicating a pressure-induced first-order phase transition has occurred. We conducted the structure refinement on the high-pressure XRD data of manganotantalite at 11.3 GPa. It is revealed that the structure of the high-pressure (HP) orthorhombic phase is assigned to the space group *Pbcn* (No. 60), same as the manganotantalite at ambient conditions. The reflections of the low-pressure (LP) and HP phases are indexed by the same indices accordingly (Figure 1). Based on the refinement results, we obtained the atomic fractional coordinates and displacement parameters of the HP phase at 11.3 GPa, revealing that the coordination numbers of Mn increase from six to eight and Ta increase from six to seven in the HP phase. However, the obtained intensities of reflections in the HP phase are not sufficient to perform high-quality structure refinements for the reason of deformation of the reflections, we have presented the refinement parameters on the HP phase at 11.3 GPa (Table S2). The recovered pure phase by decompression experiment shows that the phase transition is completely reversible upon release of pressure, indicating that this transition does not involve the breaking and remaking of a chemical bond, which is consistent with the structural characteristics of the HP phase. Our results are different from the previous report on $MnNb_2O_6$ that shows a partly reversible phase transition occurred at 12 GPa by powder X-ray diffraction, though $MnNb_2O_6$ is isostructural with Mn-$(Ta,Nb)_2O_6$.[14]

The P-V data for the two phases were fitted to a Birch−Murnaghan (BM) equation of state (EoS) as follows:[37]

$$P = \frac{3}{2}B_0\left[\left(\frac{V_0}{V}\right)^{7/3} - \left(\frac{V_0}{V}\right)^{5/3}\right] \times \left\{1 + \frac{3}{4}(B_0' - 4)\left[\left(\frac{V_0}{V}\right)^{2/3} - 1\right]\right\} \quad (1)$$

where V_0, B_0, and B_0' are the unit-cell volume, isothermal bulk modulus, and its pressure derivative at ambient conditions, respectively. In this study, the second-order BM EoS with B_0' fixed at 4 was used to fit the data to compare the elastic parameters. The yielded bulk moduli are as follows: B_0 = 149(4) GPa for the LP phase and B_0 = 188(3) GPa for the HP phase. The higher bulk modulus of the HP phase indicates that it is more difficult to compress and denser than the LP phase. The phase transition undergoes a large volume collapse of ~7.0%, also indicating a more compact crystal structure in the HP phase. Nevertheless, the bulk modulus of the LP phase of manganotantalite is consistent with that of the LP phase of $MnNb_2O_6$ (B_0 = 154 GPa) and $(Fe,Mn)(Nb,Ta)_2O_6$ solid solutions (B_0 = 146~153 GPa), compressed up to 7 GPa.[14,24]

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Abrupt changes were also observed in all of the three axes accompanied by the phase transition, including an obvious rise for the lattice parameter along the *a*-axis and reduction for those along the *b*- and *c*-axes (Figure 2). Linearized second-order BM EoS fitting with pressure derivative $M' = 12$ was used to compare the axial compressibility of the LP and HP phases, where each axial dimension is cubed and treated as volume in the BM formulation.[37] The zero-pressure axial compressibility of linear dimension *l*, defined as

$$\beta_{l0} = -(l^{-1})(\delta l/\delta P)_{P=0} \quad (2)$$

which is related to the linear modulus by

$$M_{l0} = (\beta_{l0})^{-1} \quad (3)$$

and the linear moduli and their pressure derivatives defined in this way have numerical values three times those of the corresponding volume bulk modulus,[37] that is,

$$M(i) = 3B_0(i) \quad (4)$$

$$M'(i) = 3B_0'(i) \quad (5)$$

The fitted linear moduli of *a*, *b*, and *c* are 625(15), 311(14), and 503(16) GPa for the LP phase corresponding to axial compressibility values of $\beta_a = 1.60(4) \times 10^{-3}$ GPa^{-1}, $\beta_b = 3.22(15) \times 10^{-3}$ GPa^{-1}, and $\beta_c = 1.99(6) \times 10^{-3}$ GPa^{-1} ($\beta_a/\beta_b/\beta_c$ = 1.00:2.01:1.24). It is obvious that the axial compressibility of the LP phase is highly anisotropic, with the most compressible direction along the *b*-axis; it may be correlated with the A and B cations in the columbite-type structure that might move along the *b*-axis to minimize the cation−cation repulsion during compression.[24] Based on our results, the *c*-axis is more compressible than the *a*-axis, which is in contrast to (Fe,Mn)Nb_2O_6, reflecting the effects of compositions on axial compressibility.[24] Correspondingly, the fitted linear moduli of *a*, *b*, and *c* for the HP phase are 527(9), 530(18), and 642(21) GPa with $\beta_a = 1.90(3) \times 10^{-3}$ GPa^{-1}, $\beta_b = 1.89(6) \times 10^{-3}$ GPa^{-1}, and $\beta_c = 1.56(5) \times 10^{-3}$ GPa^{-1} ($\beta_a/\beta_b/\beta_c$ = 1.00:0.99:0.82). In comparison, the *a*-axis becomes the most compressible direction in the HP phase, and the crystal anisotropy decreases significantly.

3.2. Raman Vibrational Spectroscopy. Vibrational spectroscopies are sensitive, microscopic techniques and well-suited for revealing the phase transition.[21] The acquired Raman spectrum of manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ at ambient conditions agrees well with previous reports; the modes exhibit a typical Raman feature of a columbite orthorhombic structure and have been assigned in literature and given in Table 1.[2,14] In the view of group theory analysis, Mn cations occupy the Wyckoff 4c-sites of $C_{2(y)}$ symmetry, Ta cations and three different O ions occupy 8d-sites of $C_{2(y)}$ symmetry in the orthorhombic structure with *Pbcn* space group, $MnTa_2O_6$ possesses 54 Raman-active modes ($13A_g + 14B_{1g} + 13B_{2g} + 14B_{3g}$) accordingly. The three kinds of oxygen atoms corresponding to the different Ta−O bonds in its crystal structure as follows: O_c (chaining oxygen) bonding to three Ta atoms, O_b (bridging oxygen) bonding to two Ta atoms and one Mn atom, and O_t (terminal oxygen) bonding to one Ta atom and two Mn atoms.[14] The length of Ta−O_t is the shortest bond in TaO_6 polyhedron, the strongest Raman peak at 882 cm^{-1} belongs to the Ta−O_t symmetric stretching vibration and dominates the Raman spectra in ambient conditions.[14] We conduct high-pressure Raman spectral investigation to deeply understand the pressure-induced phase transition of manganotantalite. Figure 3a displays the Raman responses of manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ as a function of pressure up to 30 GPa at room temperature in the range of 100∼1000 cm^{-1}. The behaviors of the redistribution of modes also reveal structural transition occurred at approximately 9.5 GPa, which is consistent with XRD measurements. The 11 main Raman-active modes for LP phase and 8 main Raman-active modes for HP phase observed in this studied system display significant blue shift with increasing pressure, originating from the increased strength of the interatomic interactions with reduction of bond distances. The dependence of vibrational frequencies on pressure is presented in Figure 3b. On the whole, the main Raman peaks of HP phase leap to lower frequencies compared with LP phase, suggesting an increase in average bond distance. The strongest Raman peak shows an extremely large shift to the low frequency, indicating that the bond length of Ta−O_t increases greatly in the HP phase. However, we cannot assign the other Raman peaks in the HP phase, except for the strongest peak in the present study. On release of pressure, the recovered Raman spectra also indicated the reversibility of this transition, which is in excellent agreement with the XRD results.

Table 1. Raman Frequencies ν_i of $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ at Ambient Conditions and Their Pressure-Dependence $d\nu_i/dP$ from a Linear Fitting and the Corresponding Isothermal Grüneisen Parameters γ_{iP}

ν_i	freq (cm^{-1})	symmetry	mode vibration	$d\nu_i/dP$	γ_{iP}
ν_1	200	B_{2g}	O_b−Ta/Nb−O_t bending, O_b−Ta/Nb−O_b bending	2.72	2.03
ν_2	237	A_g	O_c−Ta/Nb−O_t bending, O_t−Mn−O_t bending Mn−O_t stretching couple with O_c−Ta/Nb−O_b bending	1.17	0.74
ν_3	281	B_{1g}	Mn−O_t stretching couple with O−Ta/Nb−O bending	2.83	1.50
ν_4	305	B_{2g}	Mn−O_t stretching couple with O−Ta/Nb−O bending	0.62	0.30
ν_5	320	A_g	Mn−O_t stretching couple with O−Ta/Nb−O bending	0.49	0.23
ν_6	393	A_g	Ta/Nb−O_c stretching couple with O−Ta/Nb−O bending	3.06	1.16
ν_7	517	B_{1g}	Ta/Nb−O_c stretching	3.90	1.12
ν_8	542	A_g	Ta/Nb−O_b stretching	3.99	1.10
ν_9	638	A_g	Ta/Nb−O_b stretching	4.43	1.04
ν_{10}	840	B_{1g}	Ta/Nb−O_t stretching	3.64	0.65
ν_{11}	882	A_g	Ta/Nb−O_t stretching	3.14	0.53

Linear fitting on Raman frequencies at various pressures of both phases are performed to calculate the isothermal mode Grüneisen parameters by the following equation:

$$\gamma_{iT} = (d\nu_i/dP)K_T/\nu_{i,0} \quad (6)$$

where $d\nu_i/dP$ is the slope of each vibration shift as a function of pressure, K_T is the isothermal bulk modulus, and $\nu_{i,0}$ is each vibration frequency at initial pressure. The experimental values

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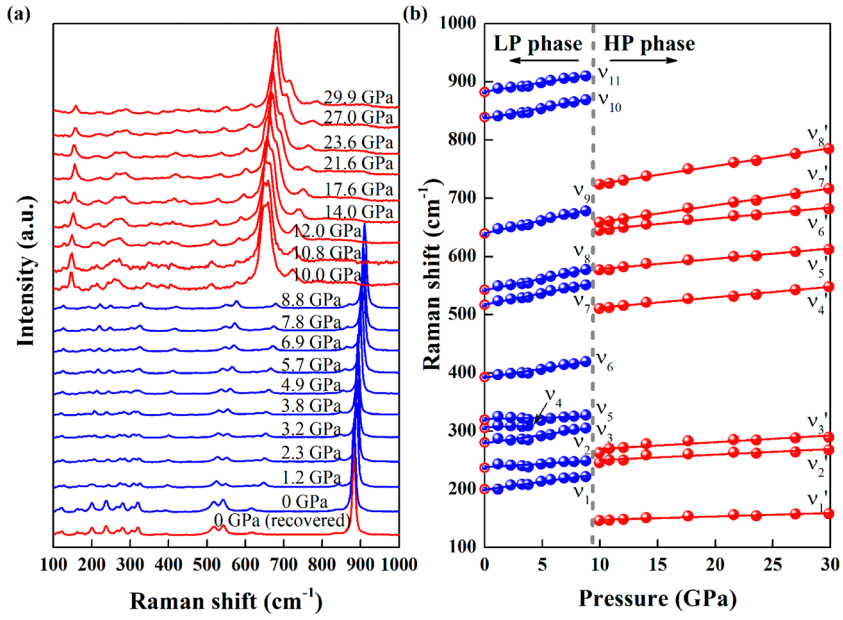


Figure 3. (a) Raman spectra of natural manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ up to 30 GPa in range of 100∼1000 cm^{-1}; (b) vibrational wavenumbers plotted as a function of pressure of the LP and HP phases, the red circles represent the vibrational wavenumbers of the recovered Raman spectrum after decompression.

of bulk moduli obtained from present XRD experiments are used to calculate the isothermal mode Grüneisen parameters, presented in Table 1 for the LP phase and Table 2 for the HP

Table 2. Raman Frequencies ν_i of the HP Phase of Manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ at 10 GPa, Their Pressure Dependence $d\nu_i/dP$ from a Linear Fitting, and the Corresponding Isothermal Grüneisen Parameters γ_{iP}

ν_i'	freq (cm^{-1})	$d\nu_i'/dP$	γ_{iP}
ν_1'	142	0.58	0.77
ν_2'	241	0.95	0.74
ν_3'	258	1.16	0.85
ν_4'	494	1.80	0.69
ν_5'	561	1.77	0.59
ν_6'	628	1.89	0.57
ν_7'	630	2.89	0.86
ν_8'	694	3.07	0.83

phase. The knowledge of these parameters reflects the sensitivity of vibrational eigenfrequencies with the change of pressure and the anharmonic effects related to thermodynamic properties. Mode softening under pressure in both phases is ruled out according to the positive values of all modes.[38] The isothermal Grüneisen parameters were determined to be 0.23∼2.03 of the LP phase. The modes observed at 200 cm^{-1} attributed to O−Ta−O bending and O−Mn−O bending vibration and modes at 320 cm^{-1} attributed to Mn−O stretching couple with O−Ta−O bending vibration show the greatest and least values of the Grüneisen parameter in the LP phase, respectively,[14] indicating the most sensitive and insensitive modes to pressure. Compared with LP phase, isothermal Grüneisen parameters of HP phase show more consistent values from 0.57 to 0.86.

We further conduct Raman experiments at HPHT conditions to explore the effects of temperature on this phase transition. Typical Raman spectra present in Figure 4a,

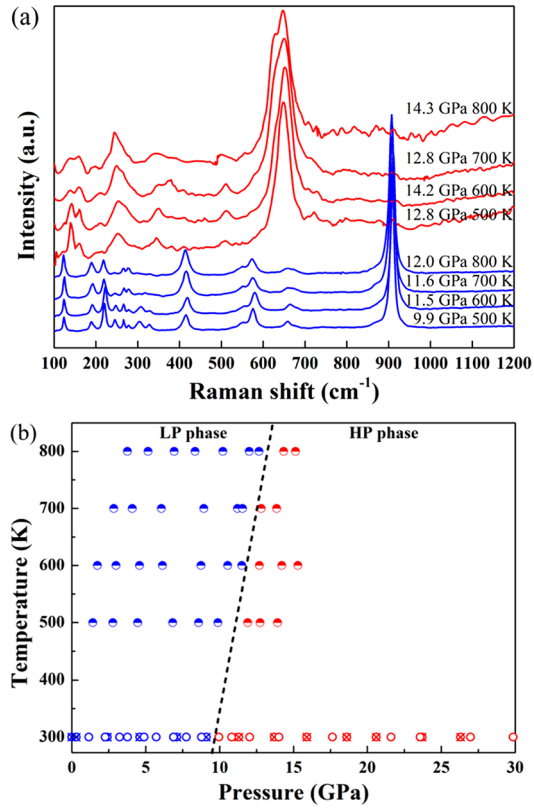


Figure 4. (a) Stacks of typical Raman spectra at selected temperatures and pressures. (b) The *P-T* phase diagram of manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ under HPHT conditions. The open and cross circles represent the results from high pressure Raman and XRD experiments at room temperature, respectively. Half up solid circles represent the results from HPHT Raman experiments.

revealing that the pressure-induced phase transition still occurred under HPHT conditions. All of the *P-T* data points derived from present HPHT Raman measurements along with

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high pressure data at room temperature were summarized in Figure 4b, corresponding to their phases. Based on these results, we obtain the phase diagram of manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ in the pressure and temperature range up to 30 GPa and 800 K. The phase transition pressure increases with temperature, indicating a positive Clapeyron slope with the value of dP/dT = 0.0073 GPa/K followed the observed *P-T* phase boundary.

3.3. Mechanism of Phase Transition. The systematic theoretical approaches using first-principle method are employed to investigate the detailed mechanism of pressure-induced phase transition behaviors on the endmember manganotantalite $MnTa_2O_6$ up to 40 GPa. Phase transition is predicted and the transition pressure of 14.2 GPa is estimated by calculating the relative enthalpy−pressure curves of two phases and finding cross point (Figure 5), columbite

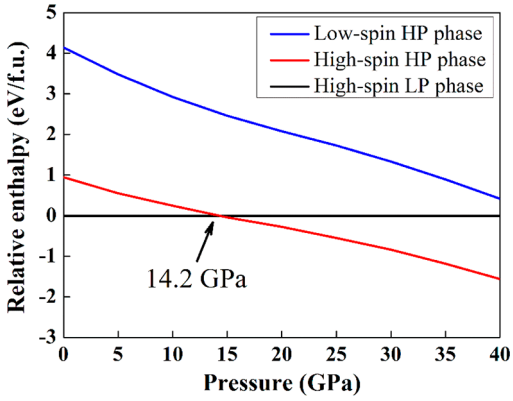


Figure 5. Static enthalpy difference as a function of pressure. The enthalpy of the LP phase with high spin Mn^{2+} was taken as a reference (black line); the red line represents the relative enthalpy of HP phase with high spin Mn^{2+} and the blue line represents the relative enthalpy of HP phase with low spin Mn^{2+}.

structure is the stable phase of $MnTa_2O_6$ at ambient conditions accordingly. The lattice parameters resulted from theoretical calculations plotted in Figure 2 and the atomic fractional coordinates presented in Table S3. The changes of lattice parameters from theoretical results are consistent with XRD measurements. The differences of lattice parameters between theoretical predicted and XRD experiments are mainly attributed to the typicality of GGA functionals theoretical calculation since the ionic radii of substituted Nb^{5+} is almost the same as that of Ta^{5+} in the natural sample.[12,39] Figure 6a shows the structural models of the LP and HP phases of $MnTa_2O_6$. The LP phase of $MnTa_2O_6$ adopts a columbite structure with Mn and Ta cations that are both octahedrally coordinated to oxygen in relatively high distorted units.[12,13] The comparative results confirm that this phase transition is accompanied by increasing the coordination oxygen atoms around Mn from six to eight and Ta from six to seven due to Bader charge transfer between Mn−O and Ta−O bonds,[40,41] which is consistent with the large volume collapse because of close packing of MnO_8 and TaO_7 polyhedral units and a reduction of the void space between polyhedra in the HP phase. Similar phase transitions related to the increase of the polyhedral coordination numbers have also been observed in ABO_4 oxides (e.g., $InVO_4$, $InTaO_4$, $InNbO_4$).[38,42,43] The spin state in $MnTa_2O_6$ has also been considered during calculations, and the individual sublattice magnetic moments are allowed to

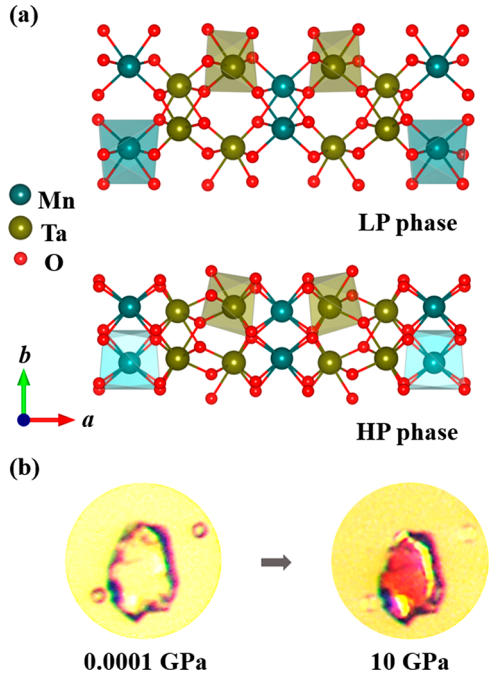


Figure 6. (a) Crystal structural features of manganotantalite $MnTa_2O_6$ (the space groups of both phases are *Pbcn*); (b) the colors of manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ at ambient conditions and 10 GPa.

relax. The magnetic moment of the Ta^{5+} cation is converged to zero after structural relaxations, indicating that the Ta^{5+} cation is in the nonmagnetic state. We calculated the relative enthalpy of high-spin and low-spin states of the Mn^{2+} cation (Figure 5), the results reveal that the spin transition of the Mn^{2+} cation is energetically unfavorable to at least 40 GPa. Therefore, the observed pressure-induced phase transition in $MnTa_2O_6$ is not accompanied by the spin crossover of the Mn^{2+} cation.

Bond lengths that varied with pressure among the two phases are plotted in Figures S2 and S3 and listed in Table S4. The average bond lengths of Mn−O and Ta−O increase along with phase transition, providing direct evidence for the leap of Raman peaks (Figure 7). The bond length of Ta−O_t increases greatly in the the HP phase from approximately 1.82 Å in the LP phase to approximately 1.95 Å in the HP phase, which is well consistent with the large shift in the strongest Raman peak based on the high-pressure Raman spectra. The larger dispersion in Mn−O bond lengths of 8-fold coordination indicates a more distorted Mn−O polyhedron in the HP phase. In contrast, we observed closer bond lengths in the Ta−O polyhedron from the LP phase to the HP phase, showing relatively regular units. The calculated distortion indexes of both Mn−O and Ta−O polyhedra presented in Figure 7 provide more direct recognition. The rapid increases in the effective coordination number of Mn and Ta, shown in Figure 7, provide comprehensive crystal chemical information on the evolution of the structure. Thus, the phase transition mechanism is different from the previously reported explanation for $MnNb_2O_6$, which is supposed to be related to the strong deformation of the NbO_6 octahedra.[14] Additionally, reasons for the difference in calculated and experimentally observed transition pressure might derive from composition differences and performing temperature of calculations at 0 K, which do not include the kinetics effects.[38] The second-order BM EOS fitting was conducted on the *P-V* data from

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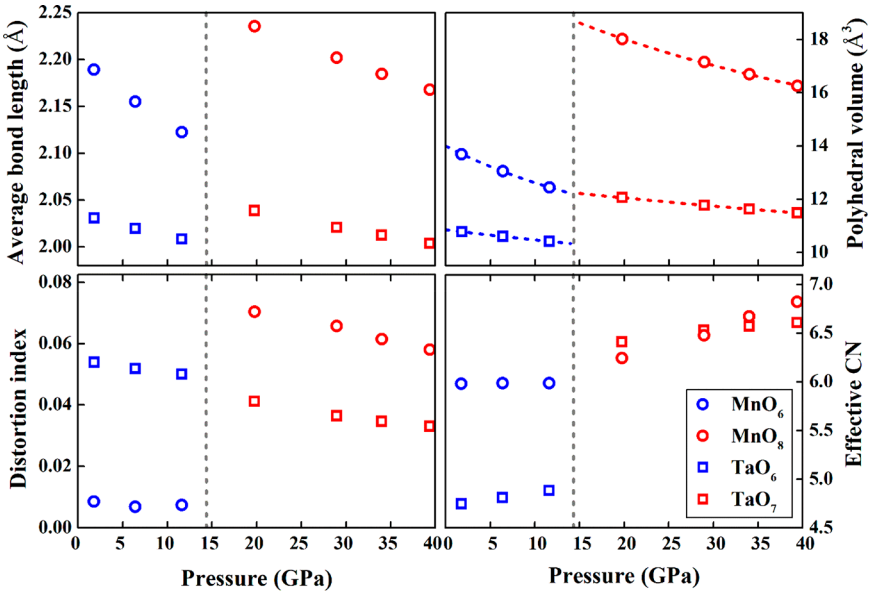


Figure 7. Average bond lengths of Mn−O and Ta−O, polyhedral volumes of Mn−O polyhedron and Ta−O polyhedron, distortion index of the polyhedron and effective coordination number (CN) in $MnTa_2O_6$ as a function of pressure based on theoretical calculations. The gray dashed line represents the phase transition pressure based on theoretical calculations. Blue and red dashed lines represent fitting curves using second-order BM EoS.

Table 3. EoS Parameters of AB_2O_6 Compounds with a Columbite-Type Structure, Determined from Present and Previous Studies

phase	V_0(Å^3)	B_0(GPa)	B_0'	methods[a]	refs
$(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ LP phase	423.4(3)	149(4)	4 fixed	exp	this study
$(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ HP phase	389.8(3)	188(3)	4 fixed	exp	this study
$MnTa_2O_6$ LP phase	439.76(4)	138.3(3)	4 fixed	theo	this study
$MnTa_2O_6$ HP phase	393.29(6)	197.0(3)	4 fixed	theo	this study
$MnNb_2O_6$ LP phase	424.2(3)	154.4(35)	4.1 fixed	exp	14
$(Fe_{0.79},Mn_{0.21})_{1.00}(Nb_{1.94},Ta_{0.06})_{2.00}O_6$ LP phase	414.56(1)	149.0(5)	4.1(2)	exp	24
$(Fe_{0.79},Mn_{0.21})_{1.00}(Nb_{1.94},Ta_{0.06})_{2.00}O_6$ LP phase	413.89(2)	153(1)	4.8(3)	exp	24
$(Fe_{0.13},Mn_{0.87})_{1.00}(Nb_{1.76},Ta_{0.24})_{2.00}O_6$ LP phase	421.18(2)	146(1)	4.9(4)	exp	24

[a]exp: experimental; theo: theoretical.

calculations, and the bulk moduli were summarized in Table 3. We obtain the polyhedral compressibility based on the calculated crystal structures. The *P-V* data of coordination polyhedra of Mn and Ta were plotted in Figure 7; second-order BM EOS bulk moduli of MnO_6 and TaO_6 polyhedral units are 78(1) GPa and 262(3) GPa for the LP phase, while those of MnO_8 and TaO_7 polyhedral units are 85(5) GPa and 292(2) GPa for the HP phase. It is indicated that the coordination polyhedra of Ta are much stiffer than those of Mn in both phases, and polyhedral units in the LP phase are more compressible than in the HP phase.

3.4. High Pressure Behavior of the Band Gap.

Strikingly, manganotantalite $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ shows a reversible drastic color change from brownish-yellow to red accompanied by phase transition presented in Figure 6b. During the phase transition, electronic distribution, and orbital interactions within the system can be changed by varying interatomic distances.[44] Therefore, such color change also suggests an alteration of electronic structure properties of manganotantalite and a narrowing of band gap. We got band gap energy of LP and HP phases based on the first-principle calculations plotted in Figure 8; an obvious decrease from 2.37 to 1.59 eV

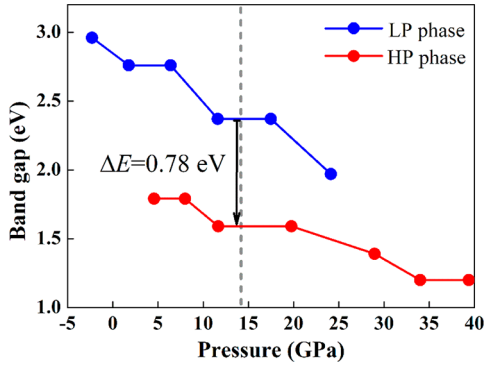


Figure 8. Changes in the band gap of $MnTa_2O_6$ are shown with respect to pressure. The gray dashed line represents the phase transition pressure based on theoretical calculations.

among the two phases at phase transition pressure is demonstrated. Compared to the gradual phase transition with a large pressure range of phase coexistence of ABO_4 compounds,[38] manganotantalite has a more sensitive and rapid phase transition. It is well-known that pressure is an important method for adjusting the properties of material. On the contrary, the special properties induced by pressure make it a

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suitable candidate as a pressure sensor. So, it is considered that manganotantalite is a potential effective pressure sensor for technological applications, such as space detection and geophysics, to monitor pressure change and warn pressure abnormality, because of its high sensitivity, fully reversible and especially drastic color change, along with pressure variation.

4. CONCLUSION

In summary, we have investigated the structural properties of manganotantalite under HPHT conditions. A pressure-induced reversible phase transition with no space group change is observed at around 9.5 GPa at room temperature, accompanied by a large volume collapse of ~7.0%. The increases in coordination oxygen numbers of Mn from six to eight and Ta from six to seven reveal the mechanism of this phase transition. The *P-T* phase diagram shows that the phase transition pressure increases with temperature with a positive Clapeyron slope of dP/dT = 0.0073 GPa/K. The narrowing of the band gap caused by a phase transition results in an intense change of color from brownish-yellow to red, which would provide a rational design strategy for the fabrication of $MnTa_2O_6$-based pressure sensor/switcher devices.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.inorgchem.0c02571.

X-ray diffraction patterns, lattice parameters, atomic fractional coordinates of $(Mn_{0.99},Fe_{0.01})_{1.00}(Ta_{1.44},Nb_{0.56})_{2.00}O_6$ at 11.3 GPa, bond lengths, and atomic fractional coordinates in $MnTa_2O_6$ by theoretical calculations (PDF)

Accession Codes

CCDC 2027882 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, by e-mailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, U.K.; fax: +44 1223 336033.

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Author Contributions

Each author has contributed to this work and all authors have approved to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

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